Unsupervised Learning for Player Role Analysis in NBA Game Data

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1 Introduction

The roles of basketball players have become increasingly fluid in modern NBA play, making traditional position labels less reflective of on-court behavior. This project investigates whether **unsupervised learning methods** can recover meaningful player roles from game-level statistics without using position labels during training.

We focus on distinguishing three broad roles: **Guard (G)**, **Forward (F)**, and **Center (C)**. Two unsupervised clustering algorithms are evaluated: **BIRCH** and **Spectral Clustering**. Their performance is compared against a **random baseline** and a **supervised Random Forest classifier**.

2 Dataset

The dataset contains NBA player game statistics for **255,915 player-game samples** and **115 total features**. Each row corresponds to a single player appearance in a game.

The features were split between numerical(105) and metadata, with the scope of not introducing noise into the training data. Metadata include player identifiers, game dates, season information, and listed starting positions. Only numerical features were used for the models. In addition, only players labeled as Guard (G), Forward (F), or Center (C) were retained.

Also, players which had less than 15 minutes played in a match were removed, due to inconsistent data. The remaining dataset contains **95,004 player-game samples**.

Final class distribution:

- Guards: 38,498
- Forwards: 38,076
- Centers: 18,430

3 Preprocessing

All models share a common preprocessing pipeline:

- **Label filtering**: only G, F, and C samples retained
- **Per-minute normalization**: all count-based statistics divided by minutes played
- **Feature scaling**: z-score normalization using StandardScaler

For Spectral Clustering, data were additionally aggregated into **player-season** representations to reduce redundancy and computational cost.

4 Feature Representations

Three distinct feature representations were evaluated.

4.1 Raw Statistical Features

This representation uses all normalized game-level statistics, including shooting, rebounding, passing, defense, and pace. It preserves the highest level of detail and dimensionality.

4.2 Role-Based Aggregated Features

The role-based representation compresses the original feature space into seven interpretable dimensions:

- Interior Presence
- Perimeter Activity
- Playmaking
- Usage / Offensive Load
- Defensive Impact
- Efficiency
- Activity and Pace

Each dimension is computed as the mean of semantically related statistics.

4.3 Behavioral and Advanced Metrics

The behavioral feature set emphasizes how players contribute, rather than total production volume. It includes:

- Shot selection distributions
- Shooting efficiency percentages
- Usage and turnover ratios
- Advanced and tracking metrics (pace, distance, impact ratings)

5 Models

5.1 BIRCH Clustering

BIRCH incrementally builds a clustering feature tree and scales well to large datasets. Hyperparameters (distance threshold and branching factor) were tuned to produce approximately three clusters.

BIRCH supports out-of-sample prediction, enabling evaluation on a held-out test set.

5.2 Spectral Clustering

Spectral Clustering constructs a k-nearest-neighbor similarity graph and partitions samples via the graph Laplacian. Due to cubic complexity, clustering was performed on a stratified subset:

- Raw features: player-season data (3,844 samples)
- Role-based features: 10,000 samples
- Behavioral features: 10,000 samples

Cosine normalization was applied prior to graph construction.

6 Hyperparameter Tuning

Although the clustering models were trained without position labels, hyperparameter selection played a critical role in determining the structure and stability of the resulting clusters. For this reason, hyperparameters were tuned using **unsupervised criteria** rather than classification performance alone.

6.1 BIRCH

For BIRCH, two hyperparameters were explored: the distance threshold and the branching factor. Rather than selecting values that maximized accuracy, we identified threshold regions that produced a **stable number of clusters** close to the expected three player roles.

In principle, BIRCH allows the number of output clusters to be fixed explicitly. However, enforcing a fixed number of clusters requires an additional global clustering pass over the CF tree. For the full game-level dataset, this step was found to be too memory-intensive and exceeded available system resources. As a result, the number of clusters could not be directly constrained to exactly three.

Therefore, across all feature representations, increasing the threshold led to a rapid reduction in the number of clusters. Thresholds were selected at points where the clustering consistently produced exactly three clusters across multiple branching factor values. This approach avoids over-fragmentation at low thresholds and cluster collapse at high thresholds.

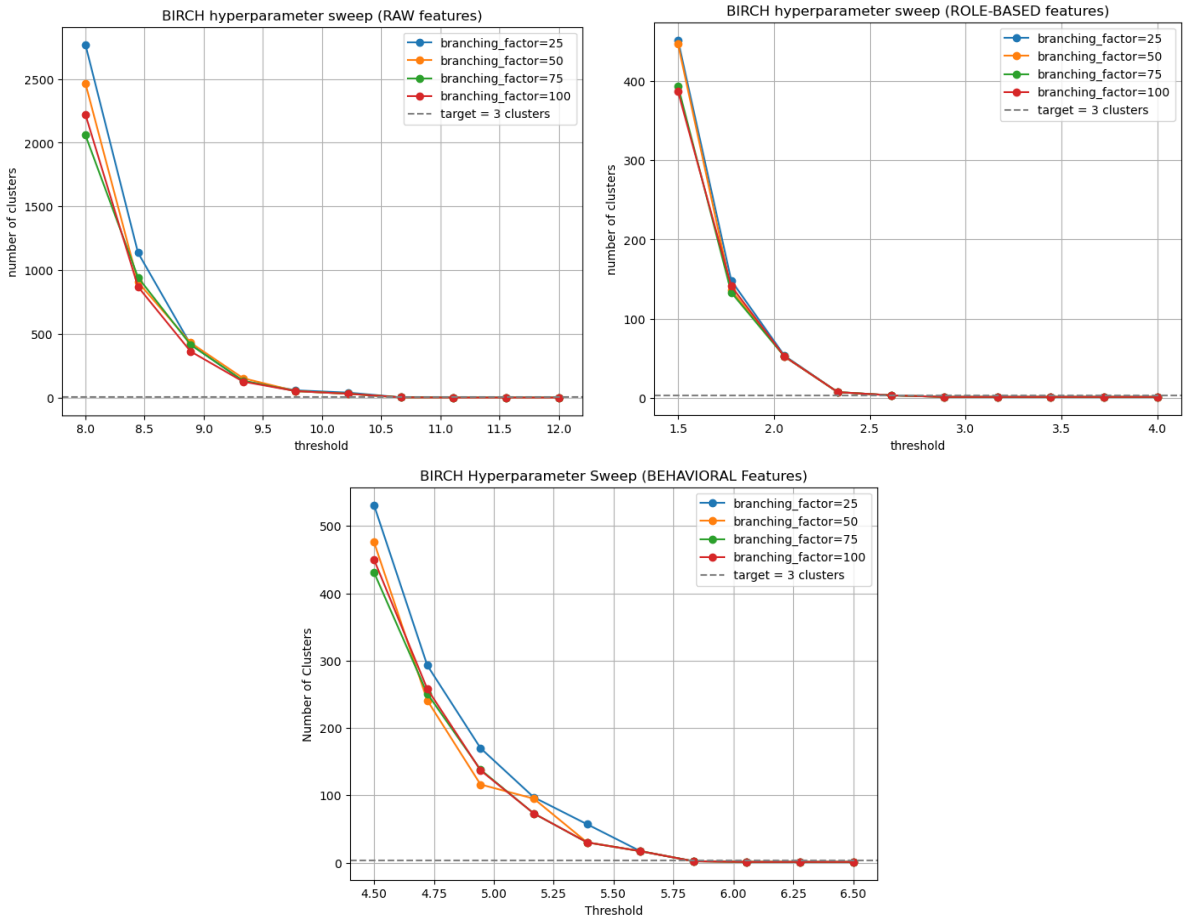

Figure 1: BIRCH hyperparameter sweeps for raw, role-based, and behavioral feature representations.

6.2 Spectral Clustering

Spectral Clustering was tuned over:

- Number of clusters ($k \in \{3, 4\}$)
- Number of nearest neighbors in the similarity graph
- Label assignment method (k-means vs. discretization)

The number of neighbors was tested over a wide range to study the transition from disconnected graphs to overly dense similarity structures. Hyperparameters were selected based on a combination of Adjusted Rand Index (ARI) and cluster stability, rather than peak accuracy alone.

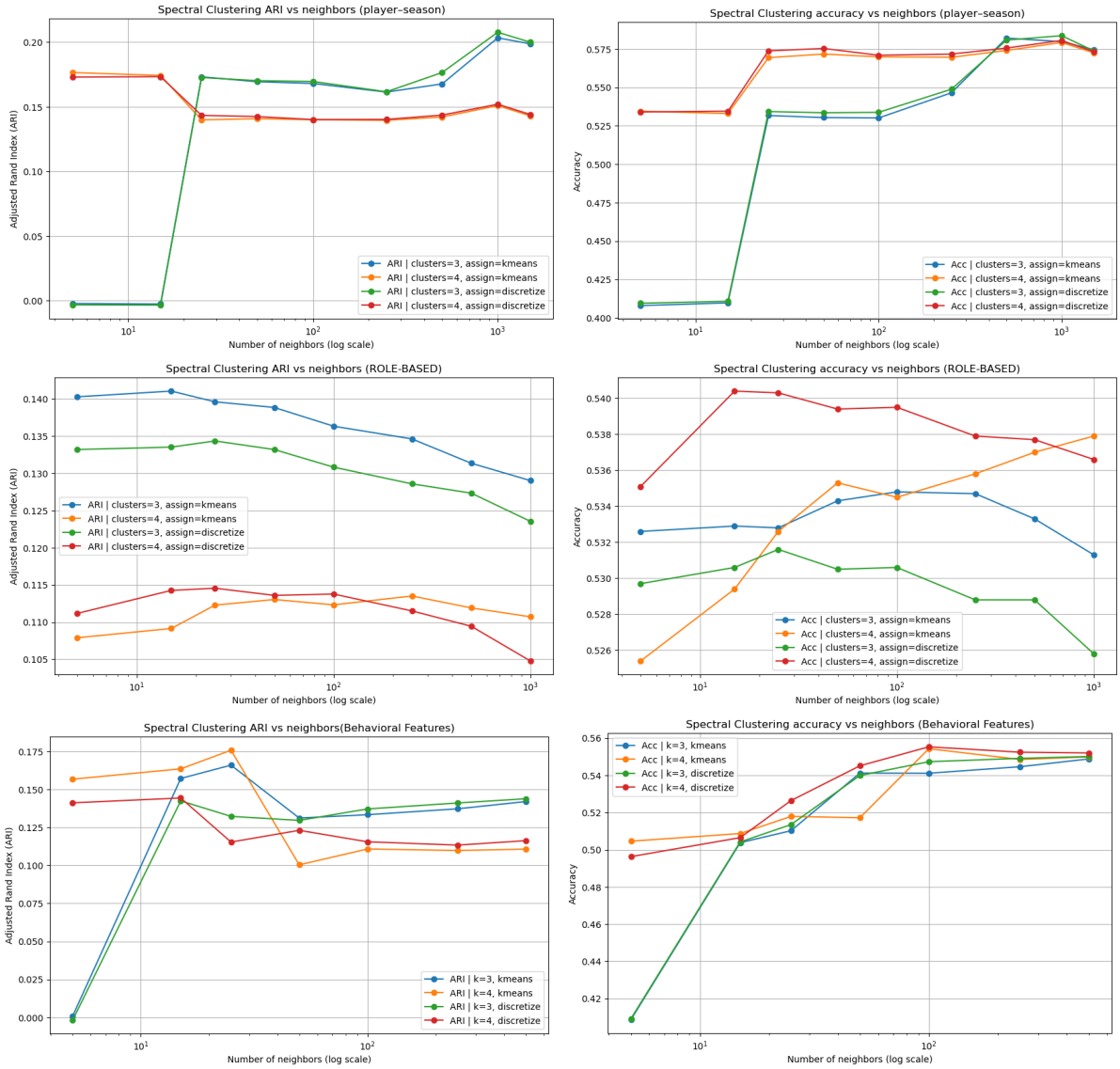

Figure 2: Spectral Clustering hyperparameter sweeps for raw, role-based, and behavioral feature representations.

7 Cluster Interpretation

Beyond numerical performance, examining the semantic meaning of the discovered clusters provides insight into what the models actually learn.

7.1 Centers as a Distinct Role

Across nearly all feature representations and models, one cluster was consistently dominated by Centers. This suggests that interior-focused statistics such as rebounding, rim protection, and paint scoring create a statistically distinct group.

Both BIRCH and Spectral Clustering were able to isolate this role with relatively low confusion, indicating that traditional Center responsibilities remain well-defined in modern NBA data.

7.2 Guard-Forward Overlap

In contrast, Guards and Forwards were frequently merged into the same cluster or split inconsistently across models. This behavior was observed most clearly in:

- Raw statistical features
- Behavioral feature representations

This overlap reflects the increasing role similarity between perimeter players, where ball handling, shooting, and defensive responsibilities are shared across positions.

7.3 Effect of Feature Representation

Role-based features produced clusters that were easier to interpret, but not necessarily more separable. Behavioral features emphasized efficiency and style, which improved supervised classification performance but reduced unsupervised separability.

These findings suggest that features optimized for prediction are not always optimal for clustering, particularly when roles form a continuum rather than discrete groups.

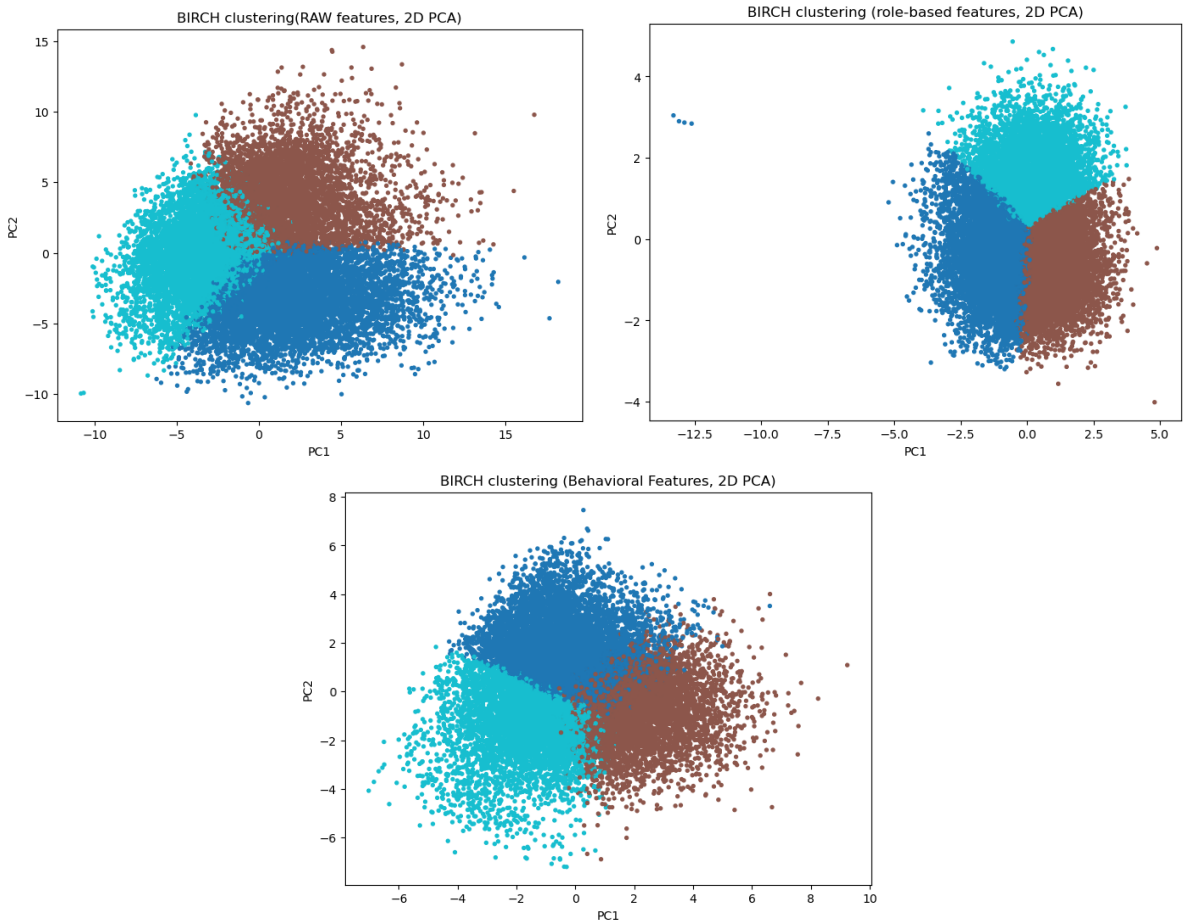

Figure 3: Two-dimensional PCA projection for all 3 feature-sets, colored by BIRCH cluster assignments.

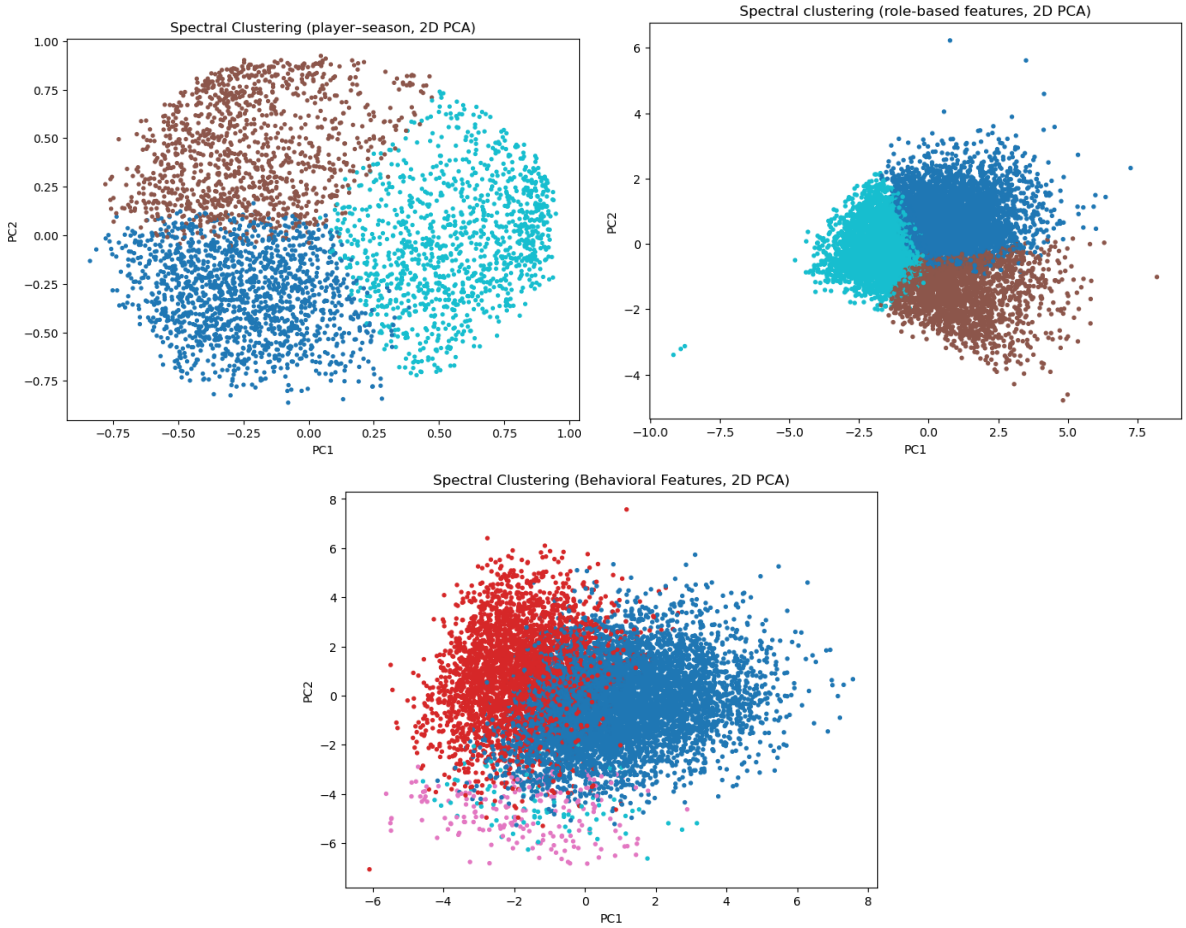

Figure 4: Two-dimensional PCA projection for all 3 feature-sets, colored by Spectral clusters assignments.

8 Evaluation Methodology

Since unsupervised models do not produce semantic labels, clusters were mapped to positions using **majority voting**. Each cluster was assigned the most frequent true position within it.

Performance metrics:

- Accuracy
- Adjusted Rand Index (ARI)
- Confusion matrices

9 Baselines

9.1 Random Baseline

Labels were sampled uniformly at random. This baseline establishes a lower bound on performance:

- Accuracy: 0.3325
- ARI: 0.0001

As expected, random predictions show no meaningful alignment with true roles.

9.2 Supervised Random Forest

A Random Forest classifier with 400 trees was trained using true position labels. Models were trained separately on each feature representation.

- **Full raw features**: Accuracy = 0.7275, ARI = 0.3450
- **Role-based features**: Accuracy = 0.6291, ARI = 0.2059
- **Behavioral features**: Accuracy = 0.6993, ARI = 0.2990

Even supervised models exhibit confusion between Guards and Forwards, highlighting the overlap between roles.

10 Results

10.1 BIRCH Clustering

- Raw features: Accuracy = 0.5328, ARI = 0.1356
- Role-based features: Accuracy = 0.5248, ARI = 0.1219
- Behavioral features: Accuracy = 0.5460, ARI = 0.1042

BIRCH produced stable three-cluster solutions but frequently merged Guards and Forwards, while Centers were more consistently isolated.

10.2 Spectral Clustering

- Player-season (raw): Accuracy = 0.5838, ARI = 0.2077
- Role-based features: Accuracy = 0.5306, ARI = 0.1308
- Behavioral features: Accuracy = 0.5179, ARI = 0.1757

Spectral Clustering achieved the highest ARI among unsupervised methods but was sensitive to hyperparameters and graph connectivity.

11 Discussion

Across all experiments, Centers were the most statistically separable role, while Guards and Forwards exhibited substantial overlap. This overlap limits the ceiling of unsupervised performance.

BIRCH demonstrated strong scalability and stability but favored dominant classes. Spectral Clustering captured more complex structure but required careful tuning and subsampling to remain computationally feasible.

Behavioral features improved interpretability and supervised performance but did not outperform raw statistics in unsupervised clustering.

12 Conclusion

This project shows that unsupervised learning can recover coarse NBA player roles beyond random chance, but with limited accuracy. Feature representation and scalability play critical roles in clustering quality.

Supervised models substantially outperform unsupervised methods, confirming that modern basketball roles are inherently fluid and overlapping.